EXPERIMENT 6

Aim

To create and compile a smart contract using Solidity, deploy it on the Ethereum Sepolia testnet through Remix IDE and MetaMask, and execute transactions to store and update data on the blockchain.

Theory

A smart contract is a self-executing program deployed on a blockchain. It contains functions and variables that define how data is stored, accessed, and modified. Smart contracts on Ethereum are written using Solidity, a high-level programming language designed for developing decentralized applications.

Remix IDE is a browser-based development environment used to write, compile, deploy, and test Solidity smart contracts. The Solidity compiler converts source code into bytecode and an Application Binary Interface (ABI), which are required for deployment and interaction with the contract.

MetaMask is a cryptocurrency wallet that connects Remix IDE to the Ethereum network and allows users to approve blockchain transactions. Sepolia is an Ethereum testnet used for testing smart contracts without spending real ETH. Test ETH is used to pay transaction gas fees.

In this experiment, a `SimpleStorage` contract was created with a variable to store an unsigned integer. The `setValue()` function modifies the stored value and generates a blockchain transaction, while the `getValue()` function reads the current value without modifying the blockchain state. An event named `ValueChanged` records the updated value in the transaction logs.

Procedure

1. Opened Remix IDE and created the file `SimpleStorage.sol`.
2. Wrote the Solidity smart contract containing `setValue()` and `getValue()` functions.
3. Compiled the contract using a compatible Solidity compiler version.
4. Connected Remix IDE to MetaMask and selected the Sepolia testnet.
5. Deployed the compiled smart contract and confirmed the deployment transaction.
6. Executed `setValue(100)` and verified the stored value using `getValue()`.
7. Executed `setValue(250)` and verified that the stored value was updated to 250.
8. Verified the transaction details and confirmation status using Sepolia Etherscan.

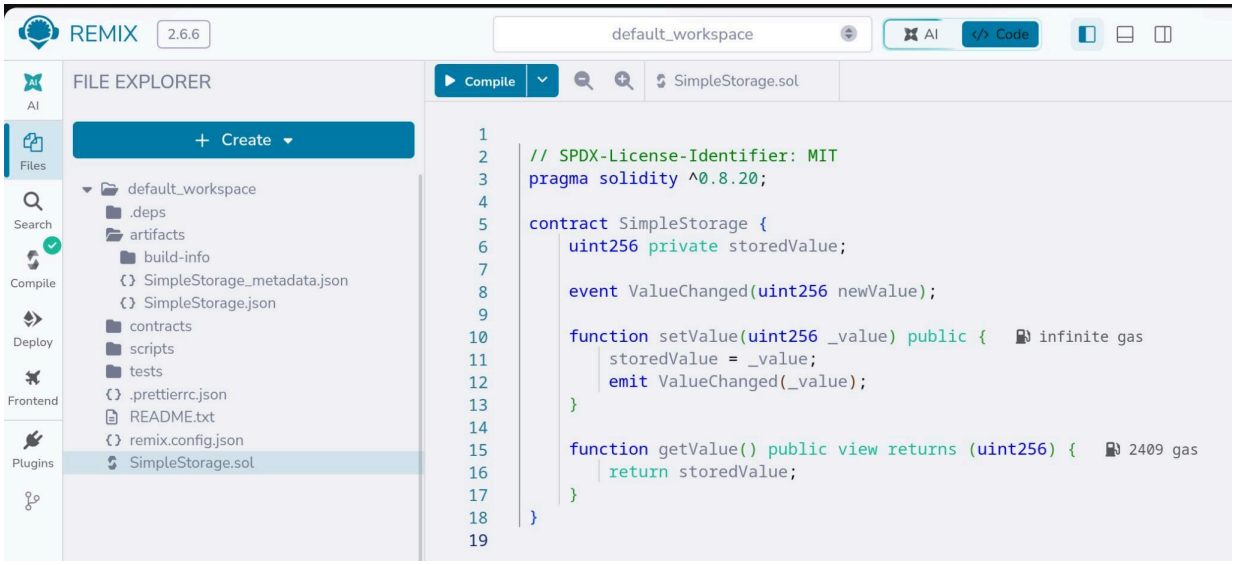
Result

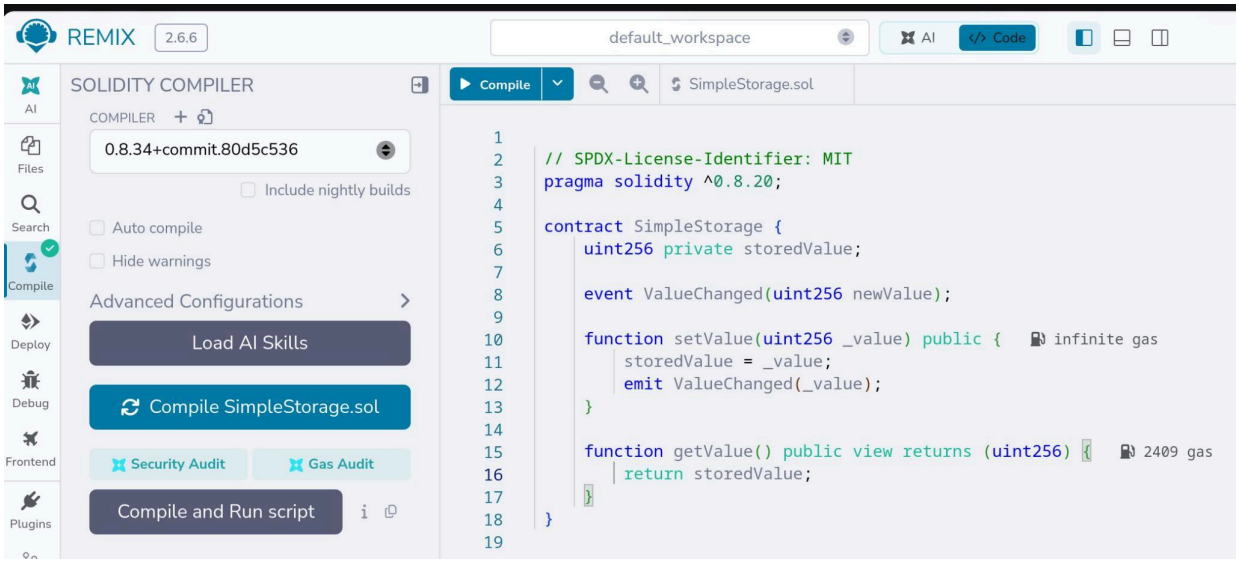
The smart contract was successfully compiled and deployed on the Sepolia testnet.

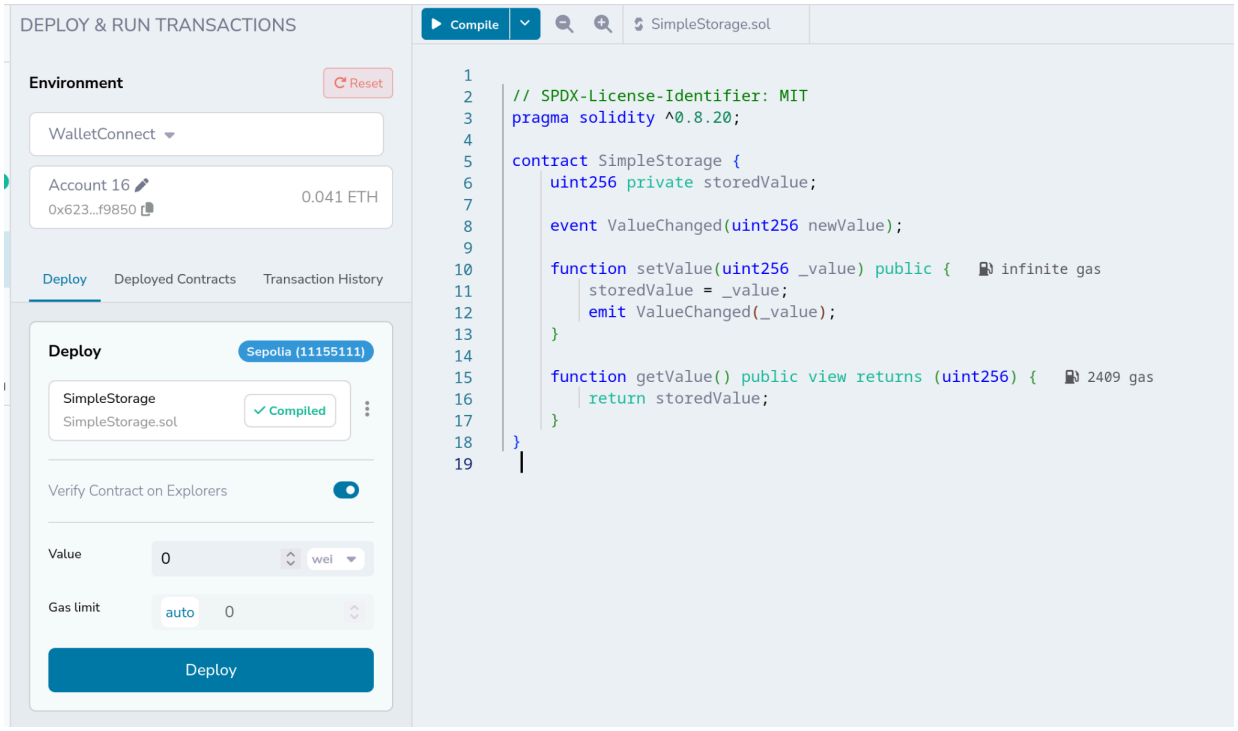
Operation	Expected result	Observed result
`setValue(100)`	Value stored as 100	Successful
`getValue()`	100	100
`setValue(250)`	Value updated to 250	Successful
`getValue()`	250	250

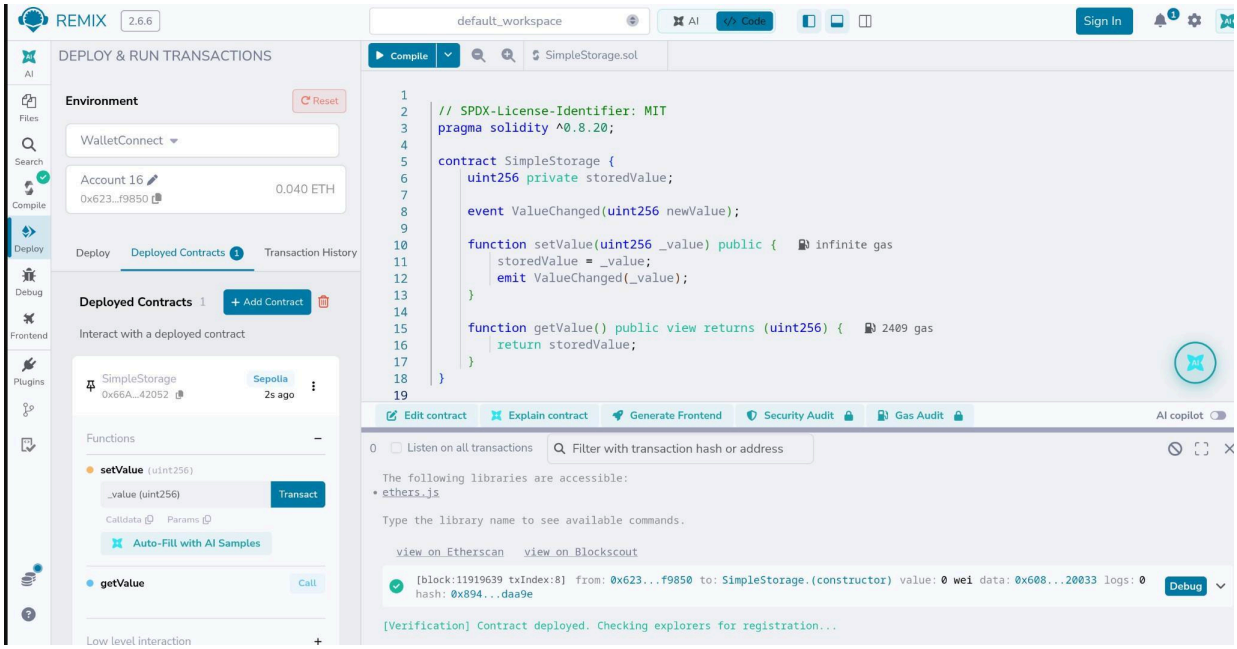
The transactions were confirmed, and the final stored value was verified as `250`.

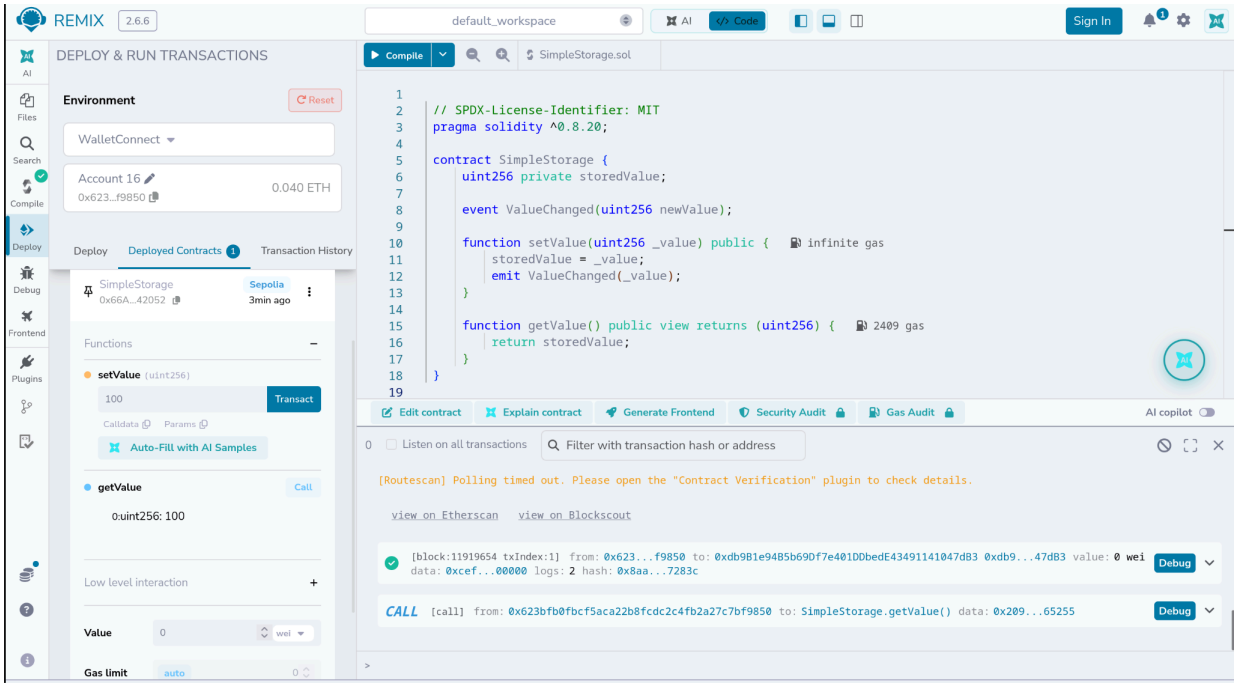
Output:

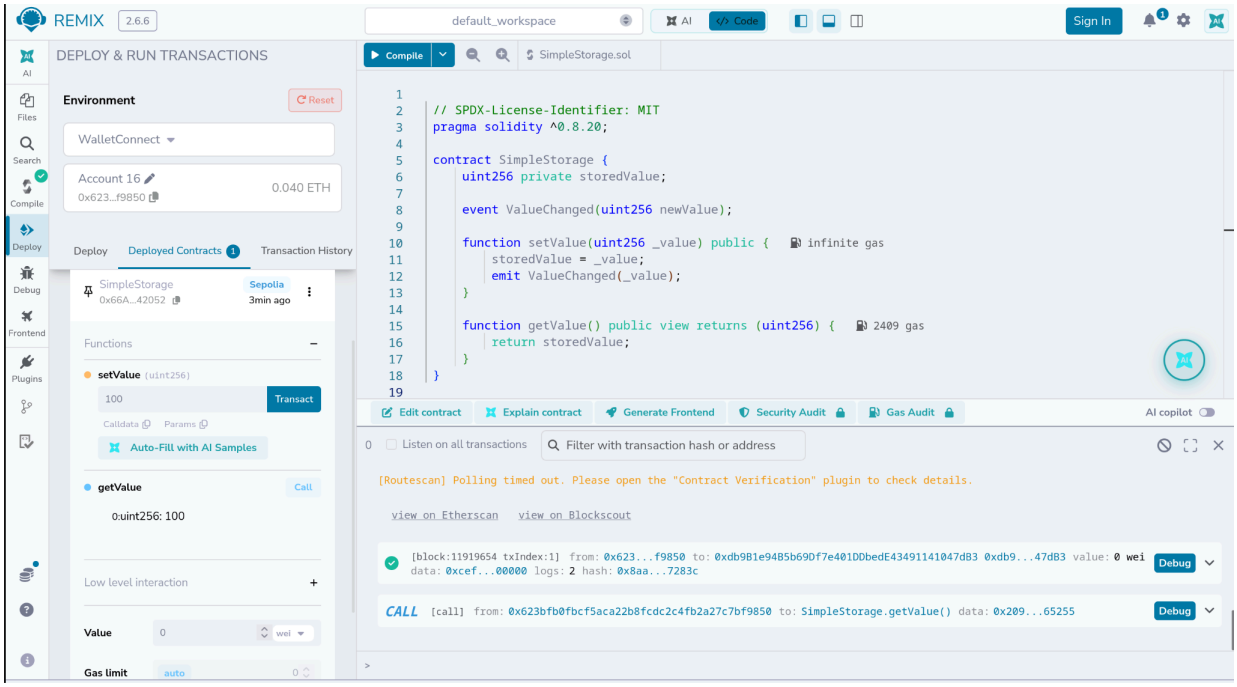


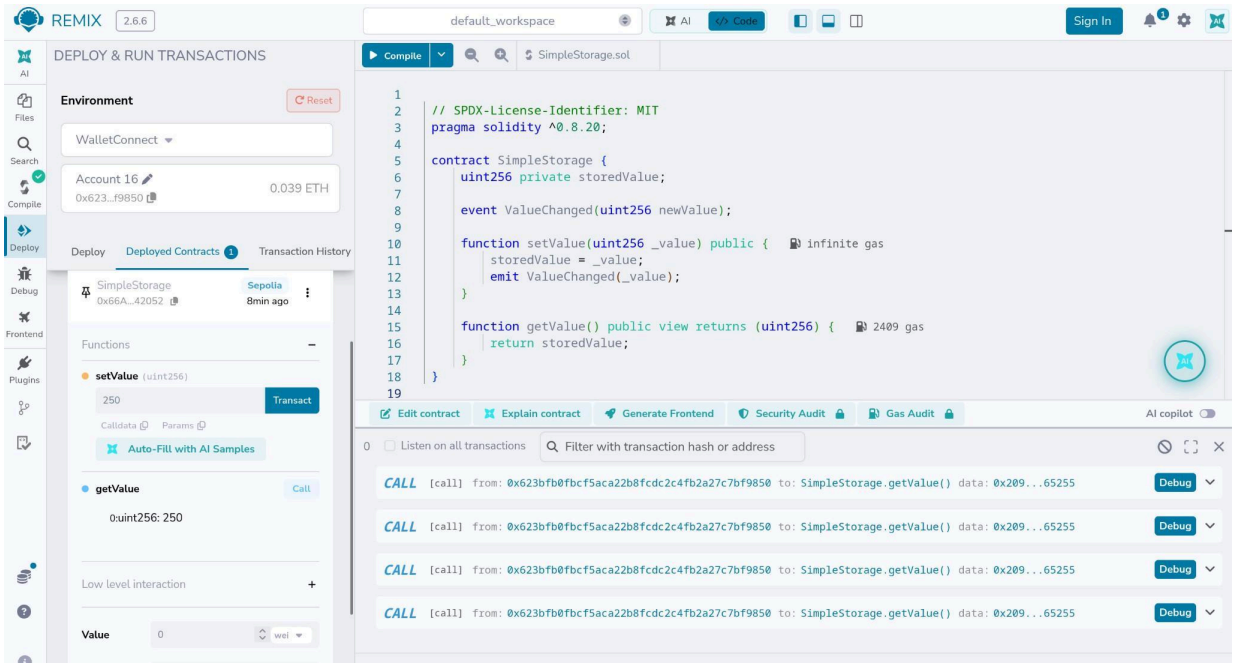

REMIX
2.6.6
default_workspace
AI
Code
SOLIDITY COMPILER
COMPILER
0.8.34+commit.80d5c536
Include nightly builds
Auto compile
Hide warnings
Advanced Configurations
Load AI Skills
Compile SimpleStorage.sol
Security Audit
Gas Audit
Compile and Run script
Compile
SimpleStorage.sol
AI
Files
Search
Compile
Deploy
Debug
Frontend
Plugins
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract SimpleStorage {
uint256 private storedValue;
event ValueChanged(uint256 newValue);
function setValue(uint256 _value) public {
infinite gas
storedValue = _value;
emit ValueChanged(_value);
}
function getValue() public view returns (uint256) {
2409 gas
return storedValue;
}
}

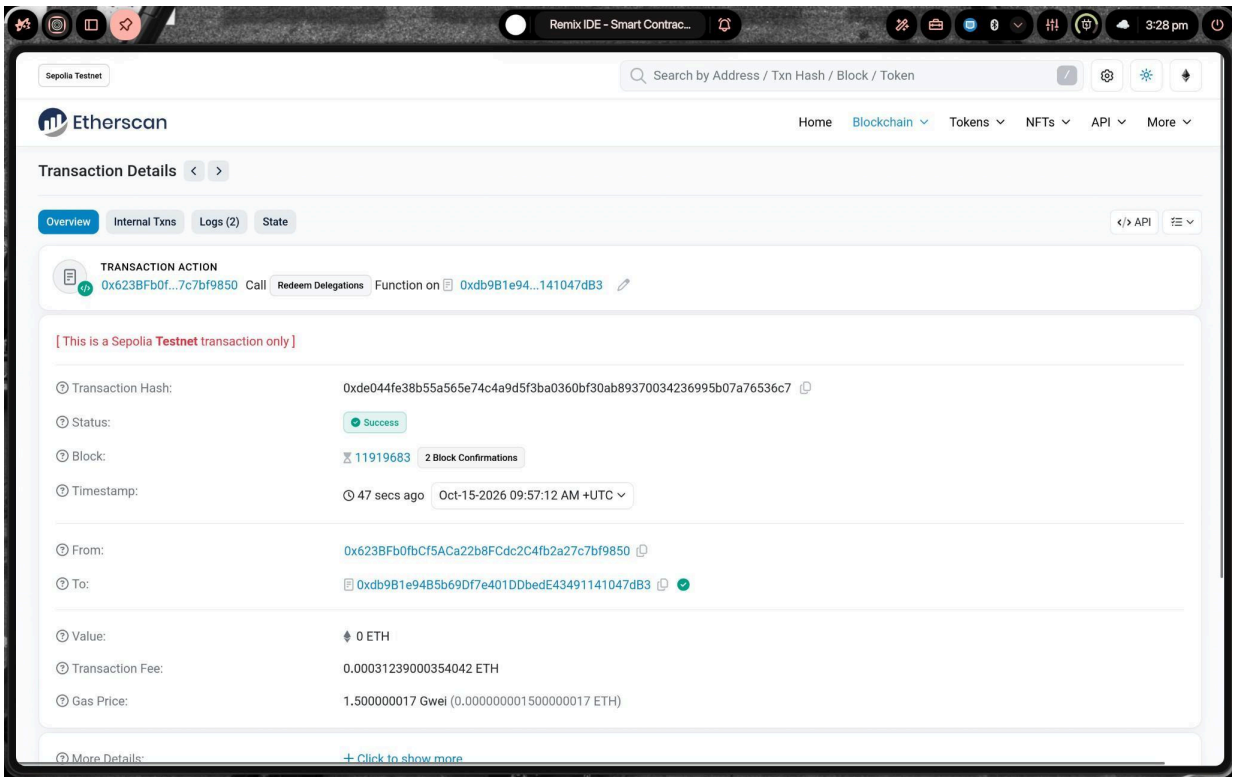

DEPLOY & RUN TRANSACTIONS
Environment
Reset
WalletConnect
Account 16
0x623...f9850
0.041 ETH
Deploy
Deployed Contracts
Transaction History
Deploy
Sepolia (11155111)
SimpleStorage
SimpleStorage.sol
Compiled
Verify Contract on Explorers
Value
0
wei
Gas limit
auto
0
Deploy
Compile
SimpleStorage.sol
// SPDX-License-Identifier: MIT
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REMIX 2.6.6
default_workspace
AI
Code
Sign In
DEPLOY & RUN TRANSACTIONS
Environment
Reset
WalletConnect
Account 16
0x623...f9850
0.040 ETH
Deploy
Deployed Contracts 1
Transaction History
Deployed Contracts 1
+ Add Contract
Interact with a deployed contract
SimpleStorage
0x66A...42052
Sepolia
2s ago
Functions
setValue (uint256)
_value (uint256)
Transact
Calldata
Params
Auto-Fill with AI Samples
getValue
Call
Low level interaction
Compile
SimpleStorage.sol
// SPDX-License-Identifier: MIT
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emit ValueChanged(_value);
}
function getValue() public view returns (uint256) {
2409 gas
return storedValue;
}
}
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
The following libraries are accessible:
ethers.js
Type the library name to see available commands.
view on Etherscan
view on Blockscout
[block:11919639 txIndex:8] from: 0x623...f9850 to: SimpleStorage.(constructor) value: 0 wei data: 0x608...20033 logs: 0 hash: 0x894...daa9e
Debug
[Verification] Contract deployed. Checking explorers for registration...


REMIX 2.6.6
default_workspace
AI
Code
Sign In
DEPLOY & RUN TRANSACTIONS
Environment
Reset
WalletConnect
Account 16
0x623...f9850
0.040 ETH
Deploy
Deployed Contracts 1
Transaction History
SimpleStorage
0x66A...42052
Sepolia
3min ago
Functions
setValue (uint256)
100
Transact
Calldata
Params
Auto-Fill with AI Samples
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Call
0:uint256: 100
Low level interaction
Value
0
wei
Gas limit
auto
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Edit contract
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Listen on all transactions
Filter with transaction hash or address
[Routescan] Polling timed out. Please open the "Contract Verification" plugin to check details.
view on Etherscan
view on Blockscout
[block:11919654 txIndex:1] from: 0x623...f9850 to: 0xdb9B1e94B5b69Df7e401DDbedE4349114104 7dB3 0xdb9...47dB3 value: 0 wei data: 0xcef...00000 logs: 2 hash: 0x8aa...7283c
Debug
CALL [call] from: 0x623bfb0fbcf5aca22b8fcdc2c4fb2a27c7bf9850 to: SimpleStorage.getValue() data: 0x209...65255
Debug


REMIX 2.6.6
default_workspace
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Code
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DEPLOY & RUN TRANSACTIONS
Environment
Reset
WalletConnect
Account 16
0x623...f9850
0.040 ETH
Deploy
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SimpleStorage
0x66A...42052
Sepolia
3min ago
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setValue (uint256)
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Calldata
Params
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Call
0:uint256: 100
Low level interaction
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Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
[Routescan] Polling timed out. Please open the "Contract Verification" plugin to check details.
view on Etherscan
view on Blockscout
[block:11919654 txIndex:1] from: 0x623...f9850 to: 0xdb9B1e94B5b69Df7e401DDbedE43491141047dB3 0xdb9...47dB3 value: 0 wei
data: 0xcef...00000 logs: 2 hash: 0x8aa...7283c
Debug
CALL [call] from: 0x623bfb0fbcf5aca22b8fcdc2c4fb2a27c7bf9850 to: SimpleStorage.getValue() data: 0x209...65255
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REMIX 2.6.6
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DEPLOY & RUN TRANSACTIONS
Environment
Reset
WalletConnect
Account 16
0x623...f9850
0.039 ETH
Deploy
Deployed Contracts 1
Transaction History
SimpleStorage
0x66A...42052
Sepolia
8min ago
Functions
setValue (uint256)
250
Transact
Calldata
Params
Auto-Fill with AI Samples
getValue
Call
0:uint256: 250
Low level interaction
Value
0
wei
Compile
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function setValue(uint256 _value) public {
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Edit contract
Explain contract
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Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
CALL [call] from: 0x623bfb0fbcf5aca22b8fcdc2c4fb2a27c7bf9850 to: SimpleStorage.getValue() data: 0x209...65255
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Debug
CALL [call] from: 0x623bfb0fbcf5aca22b8fcdc2c4fb2a27c7bf9850 to: SimpleStorage.getValue() data: 0x209...65255
Debug



Conclusion

The smart contract was successfully created using Solidity, compiled in Remix IDE, and deployed on the Ethereum Sepolia testnet using MetaMask. State-changing transactions were executed to store and update integer values on the blockchain. The results were verified using the contract's read function, demonstrating the process of smart contract deployment, transaction execution, and state verification on Ethereum.